ModelIt!

Student Activity Pack

How Do Baby Animals Look Like Their Parents?

The Science of Inherited Traits

Grade 1 | NGSS 1-LS3-1 | Patterns

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Draw a baby cat next to its parent cat. How are they alike? How are they different?

2 Circle the baby that matches its parent: show pictures of a brown dog with three puppies of different colors.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Parent Traits | Baby Traits

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Spotted Parent

Parent Traits set to spotted fur

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Striped Parent

Parent Traits set to striped fur

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do baby animals look like their parents but not exactly the same? Use evidence from your simulation to support your answers.

8 STEM Challenge: Design a Baby Animal

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YOUR ENGINEERING MISSION

Draw a baby animal based on the traits of two parent animals and explain which traits came from which parent.

THE SCENARIO

A zoo needs your help! Two parent animals just had a baby and the zookeepers need to know what the baby might look like. Look at the parent traits and draw what you think the baby will look like. Explain your thinking!

Engineering Design Thinking

1. WHAT COLOR WILL THE BABY ANIMAL BE?
2. WHAT SIZE EARS WILL IT HAVE — BIG LIKE ONE PARENT OR SMALL LIKE THE OTHER?
3. WILL THE BABY HAVE SPOTS, STRIPES, OR SOLID COLOR?
4. HOW IS THE BABY DIFFERENT FROM BOTH PARENTS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Zoologists study animals and their babies to learn how traits are passed along. They work at zoos, wildlife parks, and research centers. They earn $45,000-$75,000/year.